AA-C-00295A(GSA-FSS)
AMENDMENT - 2
August 7, 1969
SUPERSEDING
Amendment-1
July 25, 1966

AMENDMENT

TO

INTERIM FEDERAL SPECIFICATION

CHAIRS, ROTARY, STEEL, TUBULAR SPECIAL PURPOSE

This amendment was developed by Federal Supply Service, General Services Administration, Washington, D. C. 20406, based upon currently available technical information. The General Services Administration has authorized this amendment as a part of Interim Federal Specification AA-C-00295a (GSA-FSS), dated February 11, 1963.

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Paragraph 2.1 Federal specifications. Delete the following:

"TT-E-529 - Enamel, Alkyd, Semi-gloss.
UU-T-111 - Tape, Paper, Gummed (Sealing and Securing).
PPP-B-631 - Boxes, Wood, Nailed and Lock Corner.
PPP-C-843 - Cushioning Material Cellulosic.
PPP-T-45 - Tape, Gummed, Paper, Reinforced

Add the following Federal specification:

"QQ-R-571 - Rods, Welding, Copper and Nickel Alloys".

Paragraph 2.1 Federal standards. Delete the following:

Fed. Std. No. 102 - Preserving, Packaging and Packing Levels.
Fed. Std. No. 121 - Color of Painted Metal, General Office Furniture"

Paragraph 2.1 Military specifications. Delete "MIL-R-17252-Rubber, Cellular (bonded, Shredded), and substitute

"MIL-R-5001 - Rubber, Latex Foam Sponge"

Delete "MIL-L-10547 - Liners, Case, Water-proof"

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Paragraph 2.2 Other publications. Add the following:

American Society for Testing and Materials (ASTM) Standard:

ASTM 1564-64T - Tentative Methods of Test for Flexible Urethane Foam.

(Copies may be obtained upon application to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103).

Paragraph 3.2.2 Rubber. Delete and substitute the following:

3.2.2 Seat and back padding material. The chair seat and back shall be padded with one of the materials specified in 3.2.2.1, 3.2.2.2, or 3.2.2.3 except when removable seat and back upholstery covers are furnished, the seat and back pad shall be made of the material specified in 3.2.2.2 or 3.2.2.3.

FSC 7110

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Paragraph 3.2.2.1 Bonded shredded foam. Delete the entire paragraph and substitute the following

3.2.2.1 Bonded shredded cellular rubber The bonded shredded cellular rubber shall be medium condition, uncored (slab stock) The individual shredded cellular rubber particles of which the material is composed shall not be over 3/8 inch in diameter. The shredded rubber shall have a vulcanized interconnecting cellular structure with a porous surface and shall be bonded with a suitable adhesive. The density in pounds per cubic inch shall be 0.0065 minimum to 0.0085 maximum. The material shall be free from any objectionable odor, from large voids or seams The bonded material shall withstand the flexing test specified in Table II without breakdown of physical structure. The set after the specimen has remained at rest for 1 hour shall be not greater than 10 percent The bonded material shall withstand the accelerated aging test specified in Table I without suffering a change in compression resistance greater than 15 percent The padding shall be in one piece The bonded shredded cellular rubber for the seats shall be 2 inches thick before application and for the back pads shall be 1 inch thick sheet stock before application

TABLE I

Characteristic	Test Method
Flexing endurance	12111 of Federal Test Method Std No 601
Accelerated aging	12211 of Federal Test Method Std No 601

The sample unit shall be of sufficient size to conduct all tests required.

Paragraph 3.2.2.2 Foam latex. Delete the entire paragraph and substitute the following.

3.2.2.2 Slab stock latex foam rubber Slab stock latex foam rubber padding shall conform to Type II medium Grade 2 (Grade B of ASTM D 1056 of MIL-R-5901 except that compression resistance shall be determined by method A in lieu of method B as specified, and low temperature requirements shall be waived The material shall be in one piece The foam rubber shall be 2 inches thick for seat pads and one inch thick for back pads

After paragraph 3.2.2.2 add the following new paragraph

3.2.2.3 Polyurethane foam. Polyurethane foam shall be Grade US 34 and shall conform to the requirements of table II when tested as specified in the applicable sections of ASTM Method D 1564-64T The polyurethane foam for the seat pad shall be 2 inches thick and for the back pad shall be 1 inch thick

TABLE II. Physical properties of polyurethane foam

Property	Requirement
25 percent Indentation load Deflection, lbs.	34 - 4
Indentation load ratio (65 percent/25 percent , min.	1.9
Original Compression set 22 hr 158 degrees F. 90 percent Deflection percent Max	15
Steam autoclave test (Condition B percent change in 25 percent comp. load Deflect, percent Max.	-20
Comp. set 22 hr/158 degrees F 90 percent Deflect; percent Max.	20
Fatigue test	
25 percent Indentation load ratio loss, percent Max.	30
Thickness loss percent Max.	5

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Paragraph 3.2.4.1 Chair control. Delete the entire paragraph and substitute the following:

3.2.4.1 Chair controls. The chair controls shall provide for raising, lowering and revolving of the chair seat and shall also provide for tilting of the adjustable back of the Class 1 chair. Height adjustment shall be positive and non-binding and shall hold the seat at a given position until intentionally changed. The height adjustment shall withstand the test specified in 4.4.2.4. The adjustment range shall be not less than 3 inches and shall be continuous or in increments of not more than 3/4 inch. The chair controls shall be activated by means of readily accessible metal knobs or handwheels which are designed for ease of adjustment. Alternatively, seat height adjustment may be accomplished by manually lifting and lowering the seat to a positive stop without the use of other devices. All spider parts shall be constructed of steel, malleable cast iron or aluminum. The spider shall be secured to the seat at not less than four places and shall have sufficient spread to insure proper balance of the chair. Alternatively, the seat pan as specified in 3.4.1 and the spider may be a one unit permanent mold aluminum casting. When revolving action is in the spider, the spindle mounting shall be a metal tube or bracket and shall extend down and around the spindle not less than 2 inches, and shall have a seat not less than 1/4 inch thick, on which the spider shall revolve. The mounting shall be continuously welded or riveted in place making a strong well balanced, easily revolving unit. When revolving takes place in the base, the spindle shall be continuously welded or riveted to the spider and shall have a press-fit bracket or machined part extending down and around the spindle not less than 7/8 inch. Alternatively, when revolving takes place in the base, the spindle shall be pressed fit into a casting which extends down and around the spindle not less than 3/4 inch. The end of the spindle shall extend outside the casting not less than 1/16 inch and shall be deformed to prevent removal. The spindle shall be a steel rod not less than one inch in diameter. The back tilt adjustment for Class 1 chairs shall be a spring tension or rubber torsion tilt unit to control and insure proper tilt tension of the back. The back tilt tension adjustment bolts shall not be supported in the seat pan but the support must be a part of the chair control. The tilt shall have a positive safety stop which shall prevent the chair back from tilting more than 17 degrees from the initial forward position and there be any failure or breakage of the tilt unit. The spring shall be properly tempered and be of sufficient size and number of coils so that when the tension is adjusted to the normal position and the back tilted against the stop, the spring will not have reached full compression.

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Paragraph 3.2.5 Enamel. Delete the entire paragraph and substitute the following

3.2.5 Finishing materials, metal surfaces

3.2.5.1 Primer. The primer when used shall be compatible with the finish enamel used

3.2.5.2 Enamel. Enamel for the finishing shall be of the baking type in accordance with manufacturers' commercial product. The color shall be as specified in 3.2.2.1

Paragraph 3.3.1 Design. At the end of the paragraph add the following

"The chairs shall withstand the tests specified in 4.4.

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Paragraph 3.3.2.3 Brazing. Delete the entire paragraph and substitute the following

3.3.2.3 Brazing. Where brazing is permitted the following conditions shall apply. When brazing is used to join component parts in lieu of welding, the brazing method shall include the use of a nickel-chrome rod conforming to Type 2, Class RCuZn - A of QQ-R-571 and shall have a tensile strength of 70,000 p.s.i. When brazing is used simply as a filler or a filler to hide a seam or joint that has been securely joined by welding, the brazing method may include the use of another rod having adequate tensile strength to assure the brazed area will not crack or fracture under shipping, moving or in-use conditions.

Paragraph 3.4.1 Seat pan. At the end of the paragraph add the following

"Alternatively, the seat pan shall be of hardwood plywood not less than 5 ply and not less than 1/2 inch thick. The edges shall be rounded and smooth around the entire perimeter of the pan. The pan shall have not less than four perforations not greater than 3/4 inch or less than 1/2 inch in diameter. The hardwood seat pan shall be equipped with threaded steel inserts for attachment of the spider by machine screws and lock washers."

Paragraph 3.4.2. At the end of the paragraph, add the following:

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'Alterrativel., the inner par shall be contour shaped and constructed of not less than 5-ply hardwood plywood, not less thar 5/16 inch thick The edges shall be rounded and smooth around the entire perimeter of the pan The back pad shall be cemented to the inner pan The upholstery cover shall be stapled to the inner par as specified in 3.4.5. There shall be an outer pan of not less than 18 gage (0 0478 inch) steel shaped to fit over the inner par The edges of the outer par shall be rolled inward in a closed bead formation The inner pan shall be equipped with threaded steel inserts for attachment of the outer par for mounting to back frame by not less than two mounting plate 1 1/4 x 2" t read x 3/4 inch round read hex head screws and lockwashers'

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Paragraph 3.4.5 Delete the entire paragraph and substitute the following

3.4.5 Application of upholstery The upholstery covers shall be attached to the seat and back in a manner to insure smooth surfaces and even, well rounded edges and shall be permanently attached with adhesive and spring clips to the metal seat pan If the wood seat par specified in 3.4.1 is used the upholstery covers shall be attached by commercial type upholstery tacks or staples Stapler shall be coated steel and shall be pneumatically driven the full length of the staple legs flush into the wood Upholstery cover material shall be trimmed and held on the underside of the seat so as not to be visible when the chair is in use position Alternatively, the upholstery cover shall be removable and shall be attached by using picture frame wire as a drawstring Tension to prevent loosening, slipping or rippling of seat covers shall be maintained on the wire drawstring by a device which permits easy removal of wire without damage to the device or the seat pan There shall be no loose wire ends or rough or sharp edges on the tensior device The back covers secured with picture wire shall have the wire drawn tight and secured with no ends visible. The covers shall not contact unprotected sharp or hard edges of the seat or back pan. The method of attachment shall be completely free from hazardous surfaces or devices and shall prevent slippage and excessive wear of the covers after continual heavy use All seams shall be type 301 or type 401 stitching as specified in Federal Standard 751, all other edges shall be either hemmed or folded to double thickness at securing areas

Paragraph 3.5.2.3 Seat, 2nd line Delete The molded shredded foam cushion as specified in 3.2.2.1 or foam latex as specified in 3.2.2.2 and substitute the following

'The bonded shredded cellular rubber specified in 3.2.2.1 or slab stock latex foam rubber specified in 3.2.2.2, or the polyurethane foam specified in 3.2.2.3 when applicable.

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Paragraph 3.5.2.4 Back, 2nd line Delete 'The shredded foam back pad as specified in 3.2.2.1 or foam latex as specified in 3.2.2.2 and substitute the following

'The bonded shredded cellular rubber specified in 3.2.2.1 or slab stock latex foam rubber specified in 3.2.2.2, or the polyurethane foam specified in 3.2.2.3 when applicable

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Paragraph 3.6.1. Delete entire paragraph and substitute following

3.6.1 Pretreatment All ferrous metal parts of the chair except the spindle shall be prepared for finishing in accordance with one of the types of treatment specified in TT-C-490

Paragraph 3.6.2 Delete entire paragraph and substitute following

3.6.2 Enameling All ferrous metal parts of the chairs except the spindle treated as specified in 3.6.1 shall be coated with the enamel specified in 3.2 and baked using the time-temperature cycle recommended by the supplier of the enamel The dry film thickness shall be not less than 1 mil The finished surface shall be smooth uniform without runs, wrinkles, grit areas of thin film or no film and shall not show separation of color Finish panels shall withstand the test specified in 4.4.3

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Paragraph 4.2.2.1 Visual examination Delete "Table I Classification of Defects and substitute the following

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TABLE I. Classification of Defects

Examination	Defect	Classification Major	Minor
Finish enamel	Wrong color or type	X	
	Not uniform color		X
	Peeling, blistering, flakiness, or poor adhesion	X	
	Stain, dirt, overspray, foreign matter embedded		X
	Runs, wrinkles or sags		X
	Scratches, abrasions, chips or dents		X
	Area of no coating	X	
	Area of thin coating		X
Hardware	Missing, peeled, blistered, tarnished		X
	Casters not type specified	X	
	Bolts, nuts and lockwashers not plated		X
Workmanship:	Parts missing or damaged affecting use		X
	Welds missing or not sound (broken, cracked, burnt through, poor fusion etc	X	
	Welds not ground smooth, protrudes or pits not filled (outside only).		X
	Spot welds not properly spaced or not of required number		X
	Sharp edges or projection that may cause injury or tear clothing	X	
	Dimensions not within tolerance		X
	Tubes where bend not true and even	X	
	Covers not artificial leather or fabric as specified	X	
	Rubber bumpers under covers missing		X
	Base tubes not size or thickness specified	X	
	Back frame does not provide for adjustment horizontally and vertically	X	
	Adjustment of seat and back height not as specified or easily made		X
	Class 1 chairs without back tilt adjustment	X	

PART 3

After paragraph 4.4.2.3 add a new paragraph as follows

4.4.2.4 Height adjustment test. The chair shall be adjusted to a medium position (about 1-1/2 inches from lowest adjustment position) and a reference line shall be established to provide a measurement from floor to bottom of seat frame or seat bar. The measurement shall be recorded. A test load of 250 pounds shall be placed on the seat and shall remain in place for a minimum of 5 hours. The test load shall be removed. A measurement shall then be taken. Deflection from the reference line shall not exceed 1/2 inch.

Section 5 - Preparation for delivery. Delete the entire section and substitute the following:

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A, B or C as specified (see 6.2.1

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5.1.1 Level A. Unless otherwise specified (see 6.1) each chair shall be packaged assembled in the lowest rotary position. Casters shall be removed, packaged in a cotton drawstring bag, and the bag secured to the chair frame. Alternatively, the casters may be packaged in a cotton drawstring bag, and the bag packed in an inner fiberboard box together with the chair back rest. The legs, backrests, arm rests, and foot rings shall be cushioned, at the points of contact with the fiberboard box, with cushioning pads not less than 1/2 inch in thickness. The seats shall be covered with Kraft paper having a minimum basic weight of 60 pounds per ream (24x36-500). The cushioning pads may be interfolded paper, wood wool, wood fiber, or crepe cellulose wadding. The Kraft paper and cushioning pads shall be secured in place with gummed tape, pressure sensitive adhesive tape, or by tying with jute twine having a minimum breaking strength of 30 pounds. When specified (see 6.1) each chair shall be disassembled, provided such disassembly and assembly does not require the use of special tools. The disassembled parts shall be wrapped, cushioned, and the wrapping and cushioning secured as specified herein.

5.1.2 Level B. The chairs shall be packaged as specified in 5.1.1.

5.1.3 Level C. The chairs shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

5.2.1 Level A. Each complete chair, packaged as specified in 5.1, shall be packed in a box conforming to PPP-B-636, Box Style RSC or OSSC. The fiberboard shall be not less than 275 p.s.i. burst test. Each box shall be closed, waterproof sealed with tape and reinforced with bands of filament tape in accordance with the appendix to the box specification.

5.2.2 Level B. Each complete chair, packaged as specified in 5.1, shall be packed in a box conforming to PPP-B-636, Class domestic, Style RSC or OSSC. The fiberboard shall be not less than 275 p.s.i. burst test. Closure shall be in accordance with Method I of the appendix to the box specification.

5.2.3 Level C. The chairs shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with MIL-STD-129.

5.3.3 Special markings. In addition to the markings specified herein, arrows or prominent marks shall be placed on the sides of each shipping container to indicate the position in which the container shall be handled during shipment. The word 'UP' shall be placed at the point of the arrows in letters at least 2 inches in height.

The front panel of each shipping container shall be plainly marked as follows:

UP FURNITURE UP
FRAGILE, HANDLE
WITH CARE

The rear panel shall be plainly marked as follows:

FURNITURE
FRAGILE HANDLE WITH CARE
TOP
THIS SIDE UP

PAGE 1.

Paragraph 6.1 Definitions. Delete the entire paragraph and substitute the following:

6.1 Intended use. The chairs covered by this specification are tubular steel construction and have three base heights. The low base chairs are for secretarial and clerical use with general office steel furniture. They provide either a spring back or a tilting adjustable back.

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The intermediate base is intended for use with telephone boards, mechanical filing equipment or office machines. The high base is intended for use by draftsmen or laboratory workers.

Paragraph 6.1.1 Ordering data. Delete the following:

"(b) If shredded foam or foam latex cushions are required (see 3.2.2.1 and 3.2.2.2)
(c) If fixed or slip covers are required (see 3.4.3)

Figure 3 Back pad size. Delete "90 sq in." and substitute "87 square inches".

Figure 4 Back pad size. Delete "90 sq in." and substitute "87 square inches".

Figure 4. Measurement from floor to top of foot ring Delete "11-1/2 inches" and substitute "10 inches".

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